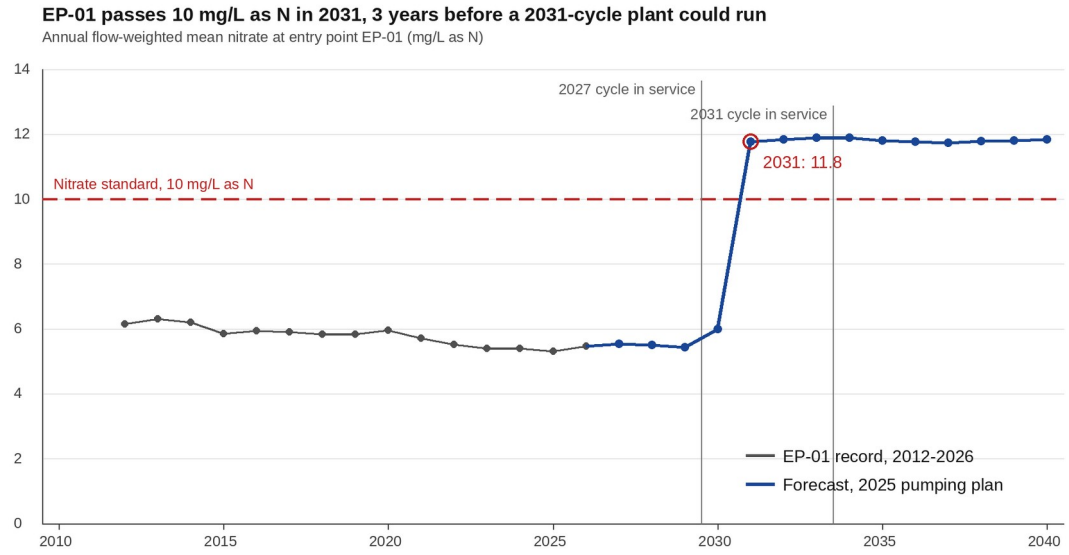
Project NR-1 belongs on the 2027 loan cycle

To the Corlinden Common Council, for its December 2026 meeting. From Chloe Griffin, general manager. 6 November 2026.

Put the nitrate removal plant on the 2027 cycle. The entry point's annual mean nitrate first reaches 10 mg/L as N in 2031 and is about 12 mg/L by 2034, so a plant funded in the 2031 cycle, in service on 1 January 2034, would come three years late. Funded in 2027 it is in service on 1 January 2030.

	2027	2029	2030	2031	2032	2034	2040
EP-01 annual mean (mg/L as N)	5.5	5.4	6.0	11.8	11.8	11.9	11.8

Source: corlinden_entry_point_nitrate_forecast.xlsx, sheet Forecast. Nitrate as N throughout; the county laboratory's results before 2004, reported as nitrate, are converted at 10 mg/L as N to 44.27 as nitrate.



The forecast blends the wells by the 2025 plan's shares (Well 3 at 8 per cent from 2027) with land cover held as it stood in 2026. Well 7 carries the most nitrate, about 16 mg/L in 2034.

Why the record looks safe

EP-01 fell from about 6.2 mg/L in 2012 to 5.3 in 2025 as water from the years of nutrient management reached the wells, and Wells 6 and 7 show no rise. Their zones of contribution take in the Sallow Ridge, about 1,099 acres of timber Hartvale Produce cleared for irrigated vegetables from 2017 to 2021, 1 to 3 years' groundwater travel from the wells. Taken at that travel time, or at the one lag of about 4.5 years that fits the wells' own record before the clearing, the ridge's nitrate should already show at Wells 6 and 7; it does not, and a model fitted to the record sets the clearing's effect to nothing.

Why it is coming

The wellhead model's travel time starts at the water table, and on the ridge the water table is 57 to 94 ft down through dry sand. The county's own wells show how long that takes: the 14 network wells whose land was put under irrigation between 1990 and 2008 each stepped up after a delay that grew with the depth to water, about 0.13 years a foot (about 1 year at 11 ft, 10 at 82 ft), while their water-table ages are under half a year. The three network wells under the 2017 to 2020 upland clearings, at 69 to 90 ft, show nothing yet, as that delay says they should. Carried to each ridge parcel's own depth, the leachate reaches the water table from 2028 and the wells around the start of 2031.

Under irrigated vegetables the network's water now carries about 21 mg/L as N, down from about 34 before nutrient management. Mixed into each well by its parcels' recharge, the ridge takes Well 6 to about 15 and Well 7 to about 16 mg/L, and the entry point past 10 in 2031.

What the forecast rests on

The delay comes from the network's matured conversions, fitted with a residual of 0.3 years; the result does not move if the half-way arrival is read as a fitted step, if the delay is taken through the origin, or if the model is left uncarried to the wells' 2019-2026 level (the first year at 10 stays 2031, the 2034 entry point 12 mg/L). The forecast

holds the land cover of 2026; further clearing in the zones would bring the date forward. The working is in corlinden_entry_point_nitrate_forecast.xlsx.